HOTEL BOOKING SYSTEM

COURSE NAME: ADVANCED PROGRAMMING (OOP) LAB

PROJECT TITLE: HOTEL BOOKING SYSTEM

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Title

HOTEL BOOKING SYSTEM
OBJECT-ORIENTED PROGRAMMING PROJECT

PROBLEM STATEMENT
MANUAL HOTEL BOOKING IS SLOW AND ERROR-PRONE.
THIS SYSTEM AUTOMATES ROOM RESERVATION AND CUSTOMER MANAGEMENT.

TARGET USERS
HOTEL STAFF AND CUSTOMERS.

System Design

High-Level Architecture
User → Main → HotelService

HotelService →
RoomManager
BookingManager
CustomerManager

Managers →
Room
Booking
Customer
Key Classes & Responsibilities

User → Abstract base class
Customer → Stores customer info
Room → Stores room details
Booking → Connects room & customer
Managers → Handle operations
HotelService → Controls workflow

Package Structure

```
Hotel/
├── Main.java
├── HotelService.java
├── Room.java
├── Booking.java
├── User.java
├── Customer.java
├── Admin.java
├── Managers/
└── CustomException.java
```

OOPS CONCEPT USED

Abstraction

Used in User abstract class

```
abstract class User {
   abstract void displayUser();
}
```

Inheritance

Customer & Admin extend User.

```
class Customer extends User
```

Polymorphism

displayUser() overridden.

```
@Override
void displayUser()
```

Design Pattern

Manager Pattern

TECHNICAL HIGHLIGHT

BOOKING VALIDATION LOGIC

```
ROOM ROOM =
ROOMMANAGER.FINDROOM(ROOMID);

IF(ROOM == NULL ||
!ROOM.ISAVAILABLE()) {

THROW NEW CUSTOMEXCEPTION(
"ROOM NOT AVAILABLE");
}
```

LIVE DEMO

VIEW AVAILABLE ROOMS

BOOK A ROOM

VIEW BOOKING HISTORY

CANCEL BOOKING

ERROR HANDLING EXAMPLE

THANK YOU